Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/09/2021
PI Name:	Allen, Antino Ph.D.		
Project Title:	Photobiomodulation to Ameliorate Neuronal Degeneration and Cognitive Decline after Mixed Field Irradiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aykin-Burns, Nukhet Ph.D. (University of Arkansas for Medical Sciences) Boerma, Marjan Ph.D. (University of Arkansas at Little Rock) Rodriguez, Analiz M.D., Ph.D. (University of Arkansas at Little Rock)		
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Task Description:	Exposure to space irradiation presents a significant risk to flight crews in the course of prolonged space exploration. One of the greatest concerns is a possibility of radiation-induced deterioration to central nervous system functions. The primary sources of radiation include Galactic Cosmic Rays (GCR) and Solar Particle Events (SPE). GCR consist of the high-energy charged nuclei of approximately 83% hydrogen, 13% helium, and 1% high mass (Z>2) and energy (HZE) particles, whereas SPE are composed of mostly energetic protons. GCR ions are able to penetrate several tens of centimeters of materials such as aluminum. As a consequence, practical levels of shielding materials will not fully absorb all space radiation, and there will be unavoidable exposures to be kept to a minimum. While previous studies have been devoted primarily to the effects of single source irradiation, there is an imperative need to utilize exposures to two or more sources to better simulate the occupational exposure that will be encountered by astronauts. Photobiomodulation (PBM) is a technique where red to near-infrared red (NIR) light (600 to 1100 nm) is exposed to neural tissues. Light in this range can be absorbed by mitochondrial cytochrome c oxidase (CCO), the crucial chromophore which mediates the effects of PBM in cells. PBM can affect neuronal function by affecting mitochondrial reduction/oxidation (redox) signaling. PBM has been demonstrated to improve cognitive function for PBM enhanced total antioxidant capacity in the hippocampus and improved cognition in a preclinical model of Alzheimer's dementia. One of the main mediators of antioxidant mechanisms is manganese superoxide dismutase (MnSOD). Multiple studies demonstrate that PBM decreases oxidative stress in mitochondria through increased MnSOD activity but it does not alter the protein expression of this mitochondrially localized antioxidant enzyme. Sirtuin 3 (Sirt3), which is the major deacetylase in mammalian mitochondria, has been linked to post-translational enzymatic activation of several metabolic enzymes, including MnSOD via deacetlyation of critical lysine residues on its active site. In this study, we propose to determine the role of Sirt3 signaling as a major regulator of increased MnSOD activity during transcranial PBM.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is currently no therapy that can reverse the effect of radiation-induced cognitive dysfunction. Clinically, photobiomodulation (PBM) is being tested as a therapy for impaired cognition following traumatic brain injury, and a randomized double-blind trial is currently enrolling participants. PBM has also improved executive functioning in a small cohort of patients with Alzheimer's dementia. These promising results indicate that PBM could be a successful therapy for cognitive impairment. While we do not know how PBM mediates beneficial effects, current preclinical studies indicate that it modulates mitochondrial redox pathways. The goal of the NASA space radiation protection program is to permit humans to live and work safely in space with an acceptable level of risk. The overarching program goal is to provide safe and effective medical countermeasures to protect astronauts of both sexes against the delayed radiation effects of galactic cosmic rays solar particle events. The proposed work will contribute to this goal by providing data about the neurological response to mixed field and protracted irradiation.
Task Progress:	Campaign 19C 6 Month old Male Mice received received .5 Gy SIM GCR (simulated galactic cosmic radiation). PBM Delivered 3 days per week for 3 months followed by cognitive testing. Y-Maze During the Y-maze, the amount of time a mouse spends exploring the novel arm of the maze compared to the familiar arm in the testing phase is indicative of its ability to retain the spatial memory encoded during the familiarization session. Sham, Sham/PBM, and Radiation/PBM animals displayed normal spatial recognition by exploring the novel arm during the testing session for a significantly longer period of time than they did the start and familiar arms: Radiation only significantly impaired short term memory they failed to distinguish the novel and familiar arms, by exploring all arms for approximately equal time, indicating a deficit in spatial memory. This data suggest that PBM preserves Short term memory tested in the Y-maze. Morris Water Maze Cognitive Testing: Latency to Platform: The Morris water maze was used to determine if PBM treatment reverses cognitive deficits after SIM GCR. Swim velocity can influence latency to the target during training sessions. The repeated-measures ANOVA revealed there was no significant effect of radiation treatment on swim speed. Further, all treatment groups were able to locate the visible platform. Animals were trained 5 days to find the hidden platform prior to the Probe trial. There was a main effect of radiation on latency to target. There was also a main effect of time on latency to target. Multiple comparison analysis showed non-PBM treated mice that received radiation took significantly longer time to locate to the platform compared to Sham, Sham/PBM, and Radiation/PBM treated animals on day 4. Probe Trial: During the probe trial, Sham treated and Sham/PBM mice showed memory retention by spending more time in the target quadrant than any other quadrant. In contrast, Radiation treated animals with Radiation only, or Radiation/PBM, did not show spatial memory retention. Overall it appears that PBM Spares Short Term Memory but not Spatial Memory. Dissociation and Flow cytometry analysis: Brain cortices were dissected, and dissociated by enzymatic digestion using the Neural Tissue Dissociation Kit An objective of this project to determine if/how those various endpoints are affected by PBM. The experimental methods chosen to accomplish these goals provide a means by which the effects of GCR and Fractionated Protons can be quantified both in terms of behavioral performance as well as specific cellular endpoints that are associated with hippocampal dependent learning and memory. We used flow cytometry to measure changes in Astrocytes (ACSA-2), Neural Precursor Cells (PSA-NCAM), Microglia (CD11b) and Oligodendrocytes (O4). We found No Significant changes in any of our measures cell counts ACSA-2. Campaign 20 B 9 Month old Male Mice received .5 Gy SIM GCR. Mice are 3 months older due to the cancelation of 20A. There appears to be gender differences in cognitive impairment dependent upon the Behavioral test performed. In the Y maze, Males are impaired; however, Females demonstrate no response to the irradiation. In the Morris Water

	maze, Males have intact Spatial memory where as Females are Impaired. Brain Hemi sections have been stained for Golgi Analysis; the other halves have been prepared for Metabolomics and Proteomic Analysis.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
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